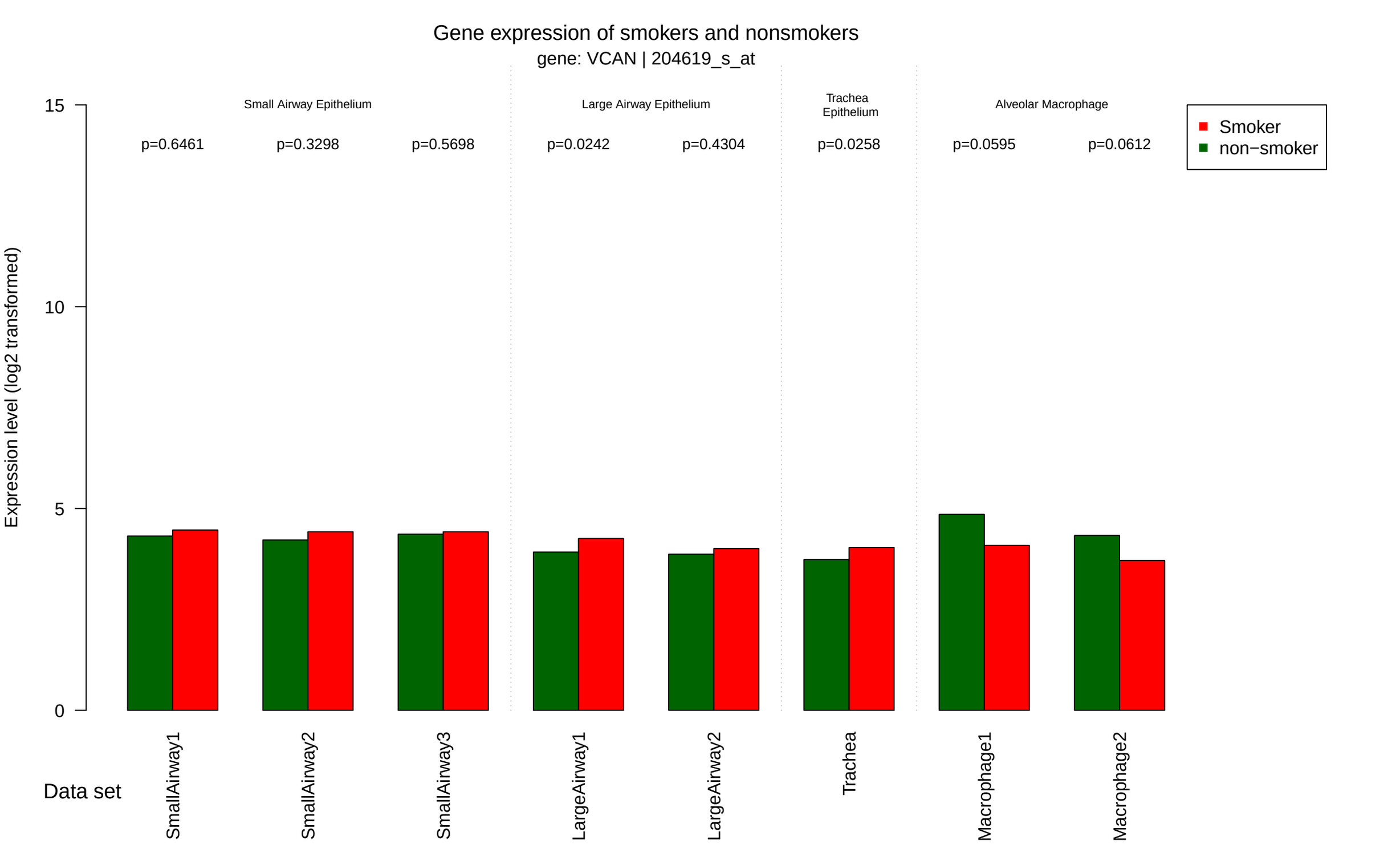
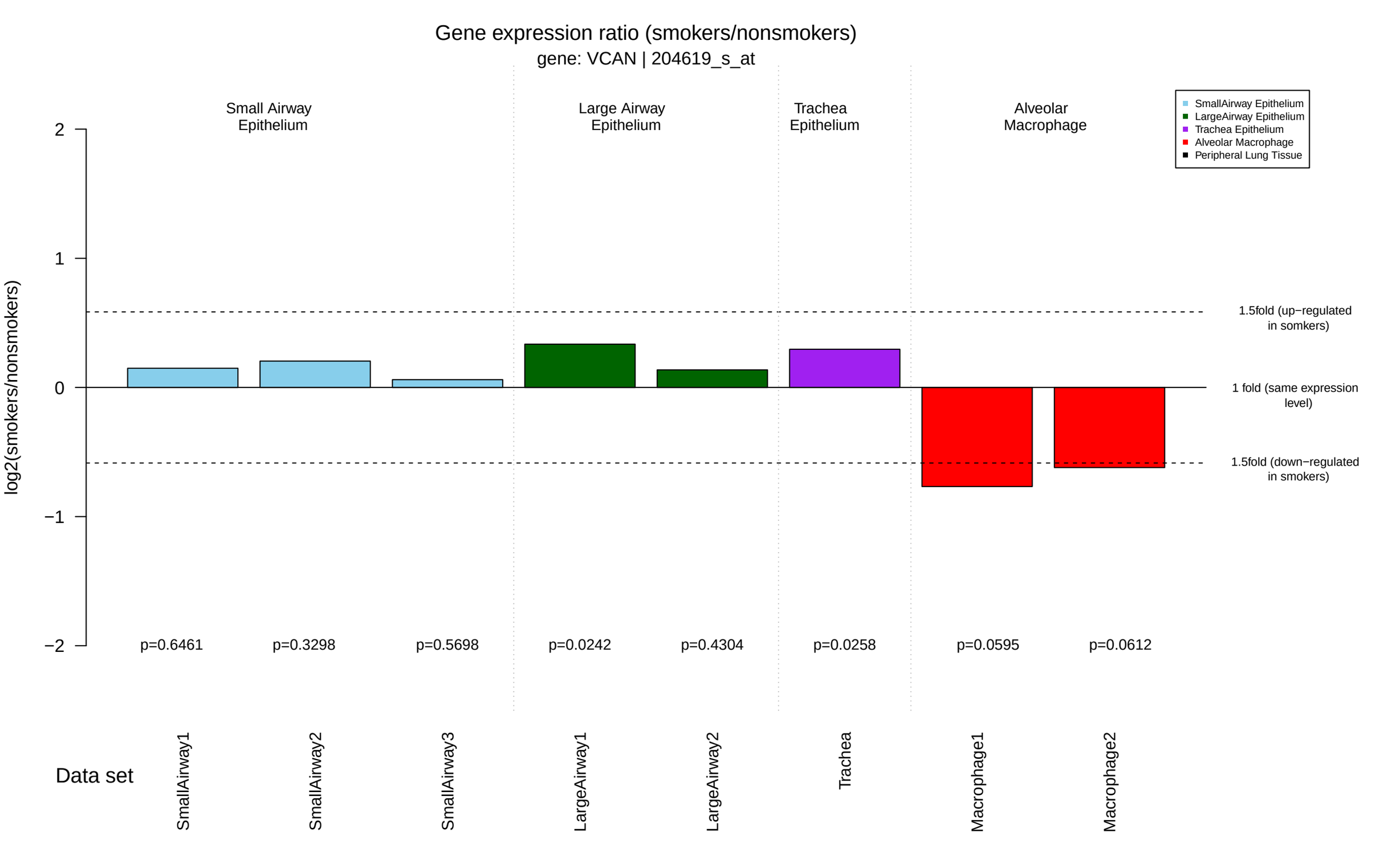
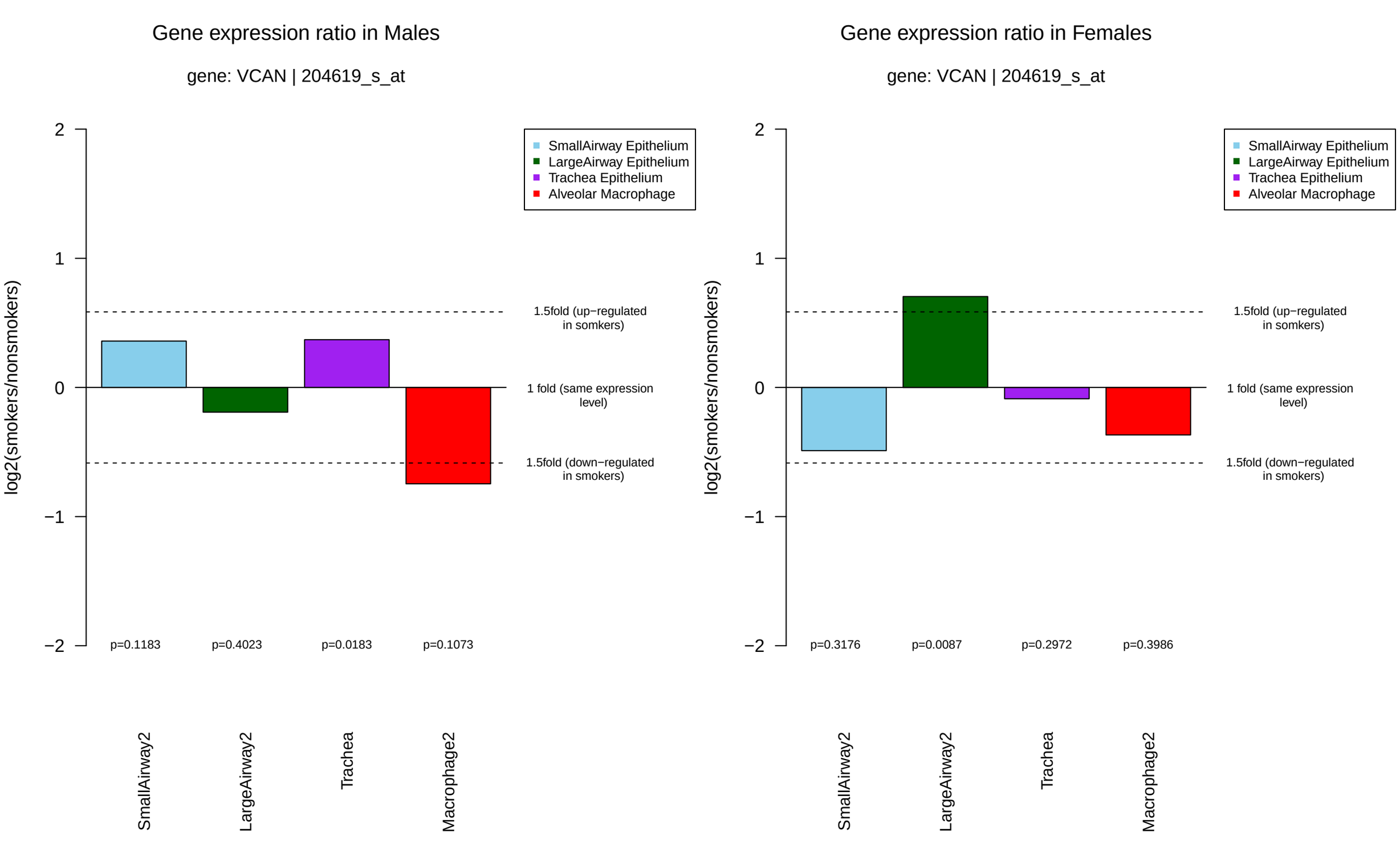
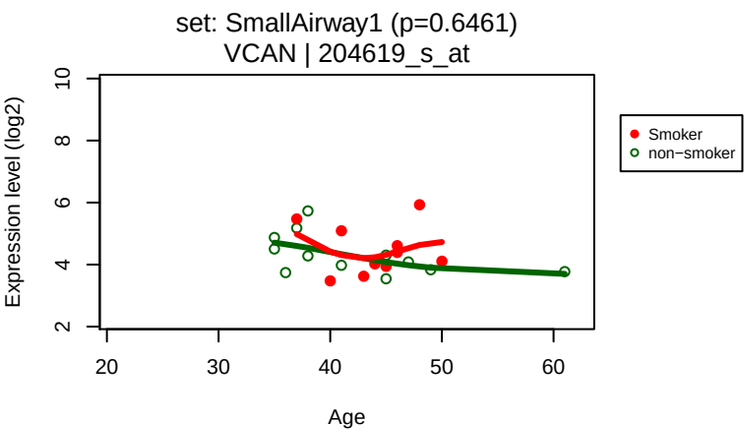
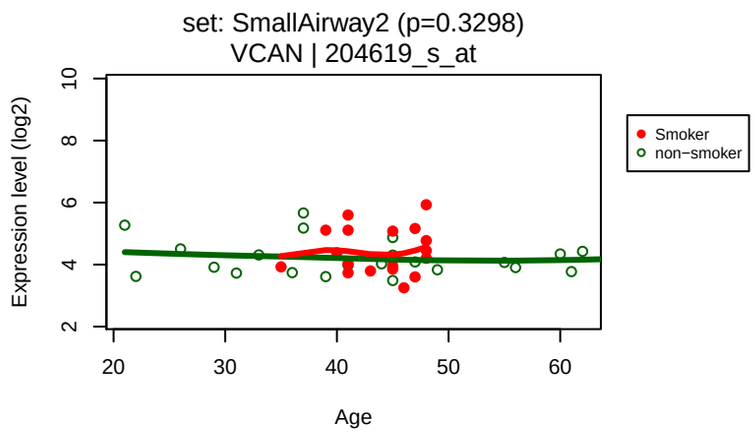
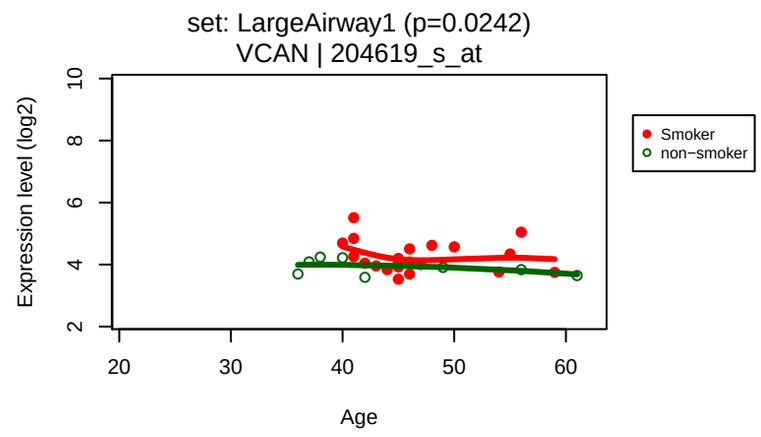

Gene expression of smokers and nonsmokers
gene: VCAN | 204619_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6461
p=0.3298
p=0.5698
p=0.0242
p=0.4304
p=0.0258
p=0.0595
p=0.0612
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

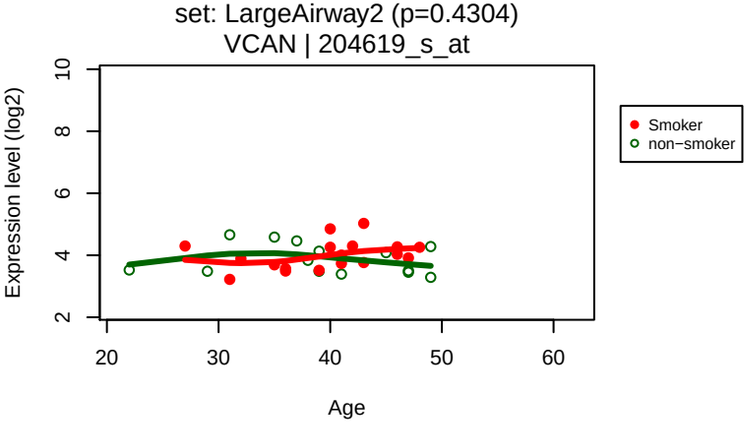

Gene expression ratio (smokers/nonsmokers)
gene: VCAN | 204619_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6461
p=0.3298
p=0.5698
p=0.0242
p=0.4304
p=0.0258
p=0.0595
p=0.0612
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

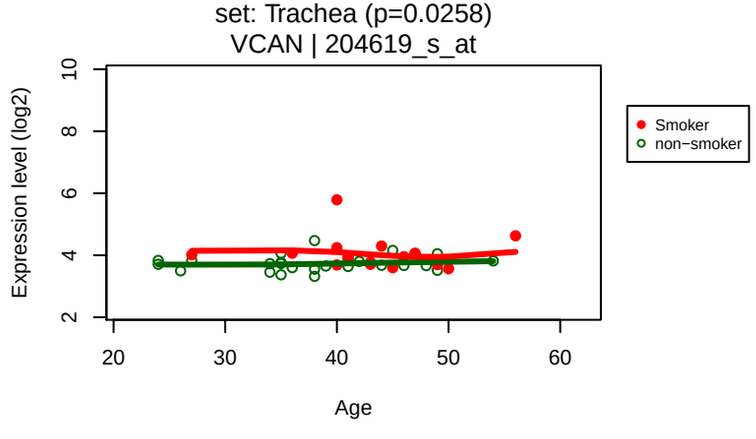

Gene expression ratio in Males
gene: VCAN | 204619_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1183
p=0.4023
p=0.0183
p=0.1073
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VCAN | 204619_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3176
p=0.0087
p=0.2972
p=0.3986
SmallAirway2
LargeAirway2
Trachea
Macrophage2

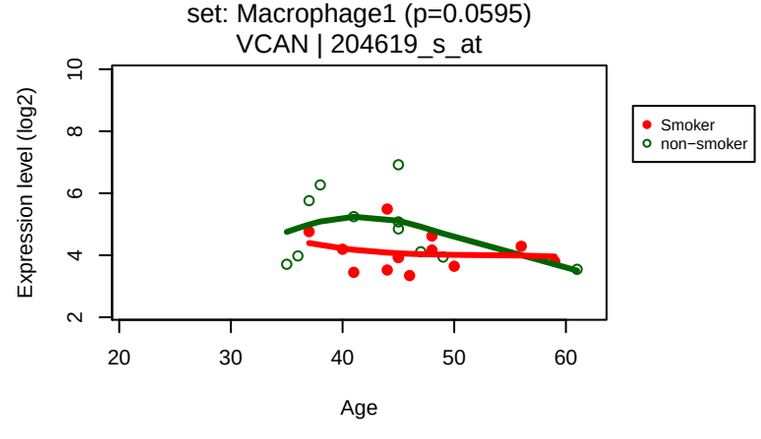

set: SmallAirway1 (p=0.6461)
VCAN | 204619_s_at
Expression level (log2)
Age
Smoker
non−smoker

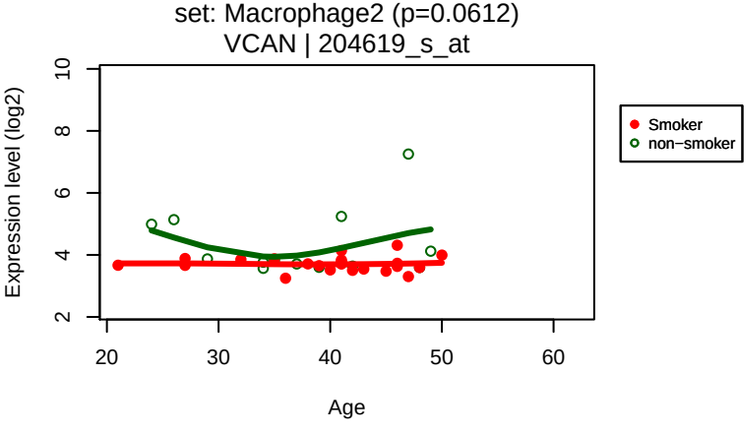

set: SmallAirway2 (p=0.3298)
VCAN | 204619_s_at
Expression level (log2)
Age
Smoker
non−smoker

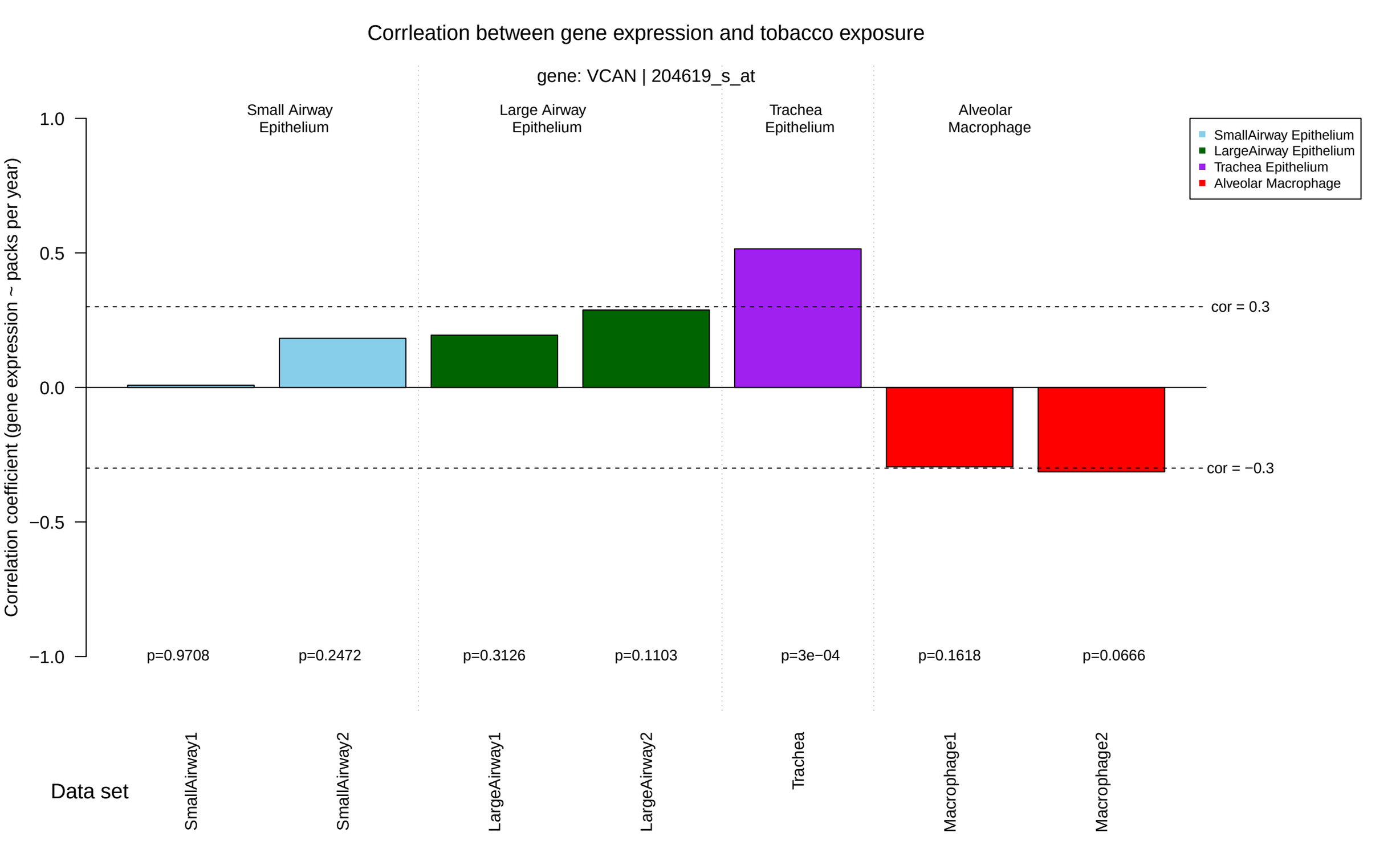

set: LargeAirway1 (p=0.0242)
VCAN | 204619_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4304)
VCAN | 204619_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0258)
VCAN | 204619_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0595)
VCAN | 204619_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0612)
VCAN | 204619_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VCAN | 204619_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9708
p=0.2472
p=0.3126
p=0.1103
p=3e−04
p=0.1618
p=0.0666
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2